

Mixture-of-Submanifolds Nonnegative Matrix Factorization for Interpretable Sparse Loading Motifs

TBD

Abstract

Nonnegative matrix factorization is widely used to represent a nonnegative data matrix as low-rank nonnegative factors and loadings. Many sparse variants of nonnegative matrix factorization encourage individual loading vectors to use only a small number of factors, but they usually do not model repeated sparsity patterns as inferential objects. We introduce mixture-of-submanifolds nonnegative matrix factorization, a probabilistic model for count data that learns recurring sparse loading motifs. Each observation is assigned, softly, to a latent motif. Each motif is represented by a small number of motif dimensions, and each dimension has a posterior distribution over the shared factor rows. This gives both clustering of observations by sparsity pattern and posterior uncertainty over the factors defining each motif. The method builds on a Poisson SuSiE model for one count vector and a matrix-factorization extension of Poisson SuSiE. We describe the model, a variational empirical-Bayes fitting procedure, and the simulation studies needed to evaluate factor recovery, motif recovery, overfitting behavior, and posterior uncertainty.

1 Introduction

Nonnegative matrix factorization (NMF) seeks a low-rank representation of a nonnegative matrix $Y \in \mathbb{N}^{N \times M}$ by writing

$$Y \approx LF,$$

where $L \in \mathbb{R}_+^{N \times K}$ contains nonnegative loadings and $F \in \mathbb{R}_+^{K \times M}$ contains nonnegative factors. In many scientific applications the rows of L are expected to be sparse: each observation should be explained by only a few latent factors. Sparse NMF methods address this by regularizing or constraining the loading matrix. However, sparsity itself is not the whole inferential target. Often we want to know whether many observations share the same kind of sparsity pattern, which factors define that pattern, and whether the evidence distinguishes one possible factor combination from another.

This paper develops MoS-NMF, a mixture-of-submanifolds model for sparse NMF loadings. The main idea is to treat recurring sparsity patterns as latent motifs. Each motif represents a small collection of factor rows that tend to appear together in the loadings. Observations are softly assigned to motifs, and each motif has posterior uncertainty over which factors define its dimensions. Thus the model is designed to answer questions such as: Which sparsity motifs are present? Which observations use each motif? Which factors define a motif? Are two nearly identical or correlated factors interchangeable in the posterior?

The statistical challenge is that sparse NMF has several sources of non-identifiability. Factor rows can be permuted, highly correlated factors can compete to explain the same counts, and an overfitted factorization can produce duplicate or redundant factors. A hard sparse loading estimate can hide this ambiguity by choosing one arbitrary support. We instead represent uncertainty with

posterior probabilities over factor identities, following the spirit of SuSiE [Wang et al., 2020]. The resulting posterior summaries are analogous to posterior inclusion probabilities, but they operate at the motif level rather than only at the observation level.

The paper has three methodological layers. Section 2 introduces Poisson SuSiE for one count vector and MF-Poisson-SuSiE for a count matrix. Section 3 introduces MoS-NMF, which pools sparse loading patterns across observations through latent motifs. Section 4 is reserved for identifiability theory. Section 5 describes the experiments needed for a JMLR-quality empirical evaluation.

2 Poisson SuSiE and MF-Poisson-SuSiE

2.1 Single-vector Poisson SuSiE

We begin with one count vector $y \in \mathbb{N}^M$ and a known nonnegative dictionary $F \in \mathbb{R}_+^{K \times M}$. Throughout this paper we use row-normalized factors,

$$F_{km} \geq 0, \quad \sum_{m=1}^M F_{km} = 1, \quad k = 1, \dots, K.$$

Poisson SuSiE represents the Poisson mean as a sparse sum of D loading dimensions:

$$y_m \mid \lambda, \gamma, F \sim \text{Poi}(\mu_m), \quad \mu_m = \sum_{d=1}^D \lambda_d F_{\gamma_d m}, \quad m = 1, \dots, M.$$

For each dimension d ,

$$\gamma_d \sim \text{Cat}(1/K, \dots, 1/K), \quad \lambda_d \mid \gamma_d \sim \text{Gam}(\alpha_d^0, \beta_d^0),$$

where the Gamma distribution is parameterized by shape and rate. The latent index γ_d chooses one factor row, and λ_d is the nonnegative loading size for dimension d .

The variational approximation factorizes over dimensions:

$$q(\lambda, \gamma) = \prod_{d=1}^D q(\gamma_d) q(\lambda_d \mid \gamma_d),$$

with

$$q(\gamma_d = k) = \bar{\gamma}_{dk}, \quad q(\lambda_d \mid \gamma_d = k) = \text{Gam}(\alpha_{dk}, \beta_{dk}).$$

We also introduce allocation variables ξ_{dm} satisfying $\xi_{dm} \geq 0$ and $\sum_{d=1}^D \xi_{dm} = 1$ for each coordinate m . They allocate the observed count y_m across the D dimensions. Jensen's inequality gives the lower bound contribution

$$y_m \log \left(\sum_{d=1}^D \lambda_d F_{\gamma_d m} \right) \geq \sum_{d=1}^D \xi_{dm} y_m \log \left(\frac{\lambda_d F_{\gamma_d m}}{\xi_{dm}} \right).$$

The resulting coordinate updates have a simple form. The allocation update is

$$\xi_{dm} \propto \exp \left[\sum_{k=1}^K \bar{\gamma}_{dk} \{ \psi(\alpha_{dk}) - \log \beta_{dk} \} + \sum_{k=1}^K \bar{\gamma}_{dk} \log F_{km} \right],$$

normalized over $d = 1, \dots, D$. The Gamma variational parameters are

$$\alpha_{dk} \leftarrow \alpha_d^0 + \sum_{m=1}^M \xi_{dm} y_m, \quad \beta_{dk} \leftarrow \beta_d^0 + \sum_{m=1}^M F_{km}.$$

The posterior probability that dimension d uses factor k is updated by a softmax over the terms in the lower bound depending on $\gamma_d = k$:

$$\bar{\gamma}_{dk} \propto \exp \left[\sum_{m=1}^M \xi_{dm} y_m \log F_{km} - \left(\alpha_d^0 + \sum_{m=1}^M \xi_{dm} y_m \right) \log \left(\beta_d^0 + \sum_{m=1}^M F_{km} \right) \right].$$

When the rows of F are normalized, the rate term is constant across k , but we keep it visible because it clarifies the role of the Gamma prior and the scale convention.

2.2 Empirical-Bayes prior update

Poisson SuSiE can use empirical Bayes to update the prior mean of each loading dimension. For dimension d , define the posterior mean and posterior mean log-size

$$\sum_{k=1}^K \bar{\gamma}_{dk} \frac{\alpha_{dk}}{\beta_{dk}}, \quad \sum_{k=1}^K \bar{\gamma}_{dk} \{\psi(\alpha_{dk}) - \log \beta_{dk}\}.$$

Maximizing the expected Gamma prior gives the rate update

$$\beta_d^0 = \frac{\alpha_d^0}{\sum_{k=1}^K \bar{\gamma}_{dk} \alpha_{dk} / \beta_{dk}},$$

and α_d^0 solves

$$\psi(\alpha_d^0) = \sum_{k=1}^K \bar{\gamma}_{dk} \{\psi(\alpha_{dk}) - \log \beta_{dk}\} + \log \beta_d^0.$$

This empirical-Bayes step is important when D is overspecified: dimensions with little evidence can have prior mean α_d^0 / β_d^0 close to zero, which is an automatic relevance determination (ARD) mechanism for selecting the effective number of dimensions.

2.3 MF-Poisson-SuSiE

MF-Poisson-SuSiE applies the same sparse Poisson representation to each row of a count matrix while learning the shared factor matrix F . For observation $i = 1, \dots, N$,

$$Y_{im} \mid \lambda_i, \gamma_i, F \sim \text{Poi} \left(\sum_{d=1}^D \lambda_{id} F_{\gamma_{id}m} \right), \quad m = 1, \dots, M.$$

Each observation has its own sparse support indicators γ_{id} and loading sizes λ_{id} . The variational posterior uses

$$q(\gamma_{id} = k) = \bar{\gamma}_{idk}, \quad q(\lambda_{id} \mid \gamma_{id} = k) = \text{Gam}(\alpha_{idk}, \beta_{idk}).$$

For fixed F , the updates are the row-wise analogues of the single-vector updates. When F is unknown, the factor update uses expected allocated counts:

$$A_{km} = \sum_{i=1}^N \sum_{d=1}^D \bar{\gamma}_{idk} \xi_{idm} Y_{im}.$$

With row-normalized factors,

$$F_{km} \leftarrow \frac{A_{km} + \epsilon}{\sum_{m'=1}^M (A_{km'} + \epsilon)}.$$

In practice the implementation initializes F with Poisson NMF, initializes the sparse posterior from the corresponding loading estimates, and damps the updates of $\bar{\gamma}$ and F . This produces a useful intermediate model: it learns the factor dictionary and observation-level sparse loading uncertainty, but it does not yet pool recurring support patterns across observations. That pooling is the purpose of MoS-NMF.

3 MoS-NMF

3.1 Generative model

MoS-NMF models repeated sparse loading motifs. The observed data are counts $Y \in \mathbb{N}^{N \times M}$. The factor matrix $F \in \mathbb{R}_+^{K \times M}$ is row-normalized as above. There are S latent motifs. We initially fit each motif with a nominal dimension D , but the effective dimension is allowed to be smaller and can vary by motif.

For motif s and dimension d , the factor identity is random:

$$\gamma_{sd} \sim \text{Cat}(\rho_{sd1}, \dots, \rho_{sdK}).$$

For observation i ,

$$Z_i \sim \text{Cat}(\pi_1, \dots, \pi_S).$$

Conditional on $Z_i = s$, the observation has one nonnegative loading size for each dimension:

$$\lambda_{id} \mid Z_i = s \sim \text{Gam}(\alpha_{sd}^0, \beta_{sd}^0), \quad d = 1, \dots, D.$$

The observed counts are

$$Y_{im} \mid Z_i = s, \lambda_i, \gamma, F \sim \text{Poi}(\mu_{im}), \quad \mu_{im} = \sum_{d=1}^D \lambda_{id} F_{\gamma_{sd}m}, \quad m = 1, \dots, M.$$

Equivalently, if L_i denotes the loading vector for observation i , then

$$L_i = \sum_{d=1}^D \lambda_{id} e_{\gamma_{Z_i d}},$$

where e_k is the k th coordinate vector in $\mathbb{R}^K$. Thus all observations assigned to motif s share the same motif-level factor identities $\gamma_{s1}, \dots, \gamma_{sD}$, but each observation has its own loading sizes.

3.2 Variational family

The variational approximation is

$$q(Z, \gamma, \lambda) = \prod_{i=1}^N q(Z_i) \prod_{s=1}^S \prod_{d=1}^D q(\gamma_{sd}) \prod_{i=1}^N \prod_{s=1}^S \prod_{d=1}^D q(\lambda_{id} \mid Z_i = s).$$

We write

$$q(Z_i = s) = \omega_{is}, \quad q(\gamma_{sd} = k) = \bar{\gamma}_{sdk},$$

and

$$q(\lambda_{id} \mid Z_i = s) = \text{Gam}(\alpha_{isd}, \beta_{sd}).$$

The rate parameter β_{sd} does not depend on i because the Poisson factor rows are shared and row-normalized; the shape parameter α_{isd} depends on the counts allocated from observation i .

For each candidate motif s , we introduce allocation variables $\xi_{isd m}$ satisfying

$$\xi_{isd m} \geq 0, \quad \sum_{d=1}^D \xi_{isd m} = 1.$$

They allocate Y_{im} across the dimensions of motif s . The implementation computes $\xi_{isd m}$ in feature blocks rather than storing the full $N \times S \times D \times M$ tensor.

3.3 Coordinate updates

The allocation update is

$$\xi_{isd m} \propto \exp \left[\psi(\alpha_{isd}) - \log \beta_{sd} + \sum_{k=1}^K \bar{\gamma}_{s d k} \log F_{k m} \right],$$

normalized over $d = 1, \dots, D$. The Gamma posterior update is

$$\alpha_{isd} \leftarrow \alpha_{sd}^0 + \sum_{m=1}^M \xi_{isd m} Y_{im},$$

and

$$\beta_{sd} \leftarrow \beta_{sd}^0 + \sum_{m=1}^M \sum_{k=1}^K \bar{\gamma}_{s d k} F_{k m}.$$

The second term is equal to one when each row of F is normalized, but we write the full expression to make the model dependence explicit.

For motif dimension (s, d) , the posterior probability of factor k is updated by

$$\bar{\gamma}_{s d k} \propto \rho_{s d k} \exp \left[\sum_{i=1}^N \omega_{is} \left\{ \sum_{m=1}^M \xi_{isd m} Y_{im} \log F_{k m} - \frac{\alpha_{isd}}{\beta_{sd}} \sum_{m=1}^M F_{k m} \right\} \right].$$

This is the central posterior uncertainty object in MoS-NMF. A concentrated vector $\bar{\gamma}_{sd1}, \dots, \bar{\gamma}_{sdK}$ means that motif dimension (s, d) has a well-identified factor. A diffuse vector means that several factor rows are plausible explanations for the same motif dimension.

The motif responsibility for observation i is updated from the component lower bound for that observation under each motif:

$$\omega_{is} \propto \pi_s \exp\{\mathcal{L}_{is}\},$$

where

$$\begin{aligned} \mathcal{L}_{is} = & \sum_{d=1}^D \sum_{m=1}^M \xi_{isd m} Y_{im} \left\{ \psi(\alpha_{isd}) - \log \beta_{sd} + \sum_{k=1}^K \bar{\gamma}_{s d k} \log F_{k m} - \log \xi_{isd m} \right\} \\ & - \sum_{d=1}^D \sum_{m=1}^M \frac{\alpha_{isd}}{\beta_{sd}} \sum_{k=1}^K \bar{\gamma}_{s d k} F_{k m} - \sum_{d=1}^D \text{KL}\{\text{Gam}(\alpha_{isd}, \beta_{sd}) \parallel \text{Gam}(\alpha_{sd}^0, \beta_{sd}^0)\}. \end{aligned}$$

The mixture weights are updated by

$$\pi_s \leftarrow \frac{1}{N} \sum_{i=1}^N \omega_{is}.$$

3.4 Learning the factors

When F is unknown, MoS-NMF updates F using the expected allocated counts

$$A_{km} = \sum_{i=1}^N \sum_{s=1}^S \sum_{d=1}^D \omega_{is} \bar{\gamma}_{sdk} \xi_{isd m} Y_{im}.$$

The row-normalized update is

$$F_{km} \leftarrow \frac{A_{km} + \epsilon}{\sum_{m'=1}^M (A_{km'} + \epsilon)}.$$

In the current implementation, F is initialized by MF-Poisson-SuSiE with learned F , and the subsequent MoS-NMF updates alternate between motif responsibilities, motif posteriors, Gamma loading posteriors, empirical-Bayes prior updates, and optionally factor updates. This initialization is important because the full problem is highly nonconvex.

3.5 Empirical-Bayes dimension selection

Each fitted motif starts with the same nominal dimension D , but the effective dimension of motif s can be smaller. MoS-NMF uses the empirical-Bayes Gamma prior parameters to diagnose dimension activity. For each motif s and dimension d , the prior mean

$$\frac{\alpha_{sd}^0}{\beta_{sd}^0}$$

estimates the loading size expected for that dimension. If this value is close to zero relative to the other dimensions in motif s , then dimension d is effectively inactive. This gives an ARD-style selected dimension

$$\hat{D}_s = \# \left\{ d : \frac{\alpha_{sd}^0 / \beta_{sd}^0}{\sum_{d'=1}^D \alpha_{sd'}^0 / \beta_{sd'}^0} \geq \tau \right\},$$

for a threshold τ .

This ARD diagnostic should be interpreted separately from the entropy of $\bar{\gamma}_{sd1}, \dots, \bar{\gamma}_{sdK}$. The ARD prior mean asks whether a dimension carries loading mass. The entropy of $\bar{\gamma}_{sd}$ asks whether the factor identity of that dimension is uncertain. Both are useful: an inactive dimension should have small loading mass, while an active but ambiguous dimension may have large loading mass and diffuse posterior factor identity.

4 Identifiability Theory

This section will contain the identifiability theory. The theory should separate at least three issues:

1. permutation and scaling conventions for the NMF factorization;
2. identifiability of motif supports given the factor matrix F ;
3. identifiability of the mixture of sparse motifs when motifs share factors or when no motif contains a single anchor factor.

The empirical sections below are designed to stress these same issues before the theory is complete.

Algorithm 1 MoS-NMF variational empirical-Bayes fitting

- 1: Initialize F , typically using MF-Poisson-SuSiE with learned factors.
 - 2: Initialize ω_{is} by clustering preliminary loading scores.
 - 3: Initialize $\bar{\gamma}_{sdk}$ from cluster-level loading patterns.
 - 4: Initialize α_{isd} , β_{sd} , π_s , and $(\alpha_{sd}^0, \beta_{sd}^0)$.
 - 5: **for** $t = 1, \dots, T$ **do**
 - 6: Update ξ_{isdm} in feature blocks.
 - 7: Update α_{isd} and β_{sd} .
 - 8: Update ω_{is} and π_s .
 - 9: Update $(\alpha_{sd}^0, \beta_{sd}^0)$ by empirical Bayes.
 - 10: Update $\bar{\gamma}_{sdk}$ for each motif dimension.
 - 11: **if** factor learning is enabled **then**
 - 12: Update F using expected allocated counts and row-normalize.
 - 13: **end if**
 - 14: Monitor the variational objective or a scheduled approximation to it.
 - 15: **end for**
 - 16: Report motif responsibilities ω , motif posteriors $\bar{\gamma}$, selected dimensions $\hat{D}_s$, and fitted factors F .
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5 Experiments Needed for a JMLR-Quality Evaluation

The empirical goal is not only to show good reconstruction error. The paper needs to show that MoS-NMF recovers meaningful sparse motifs, clusters observations by motif, learns factors when F is unknown, and quantifies uncertainty in settings where hard sparse supports are unstable.

5.1 Baselines

The main baselines should be:

1. Poisson NMF followed by clustering of normalized loading scores;
2. MF-Poisson-SuSiE followed by clustering of posterior loading scores;
3. sparse NMF baselines with tuned sparsity penalties, if available;
4. ablations of MoS-NMF with fixed F , learned F , and different initialization schemes.

The first two baselines are already implemented in the current benchmark code.

5.2 Metrics

The simulation tables should include:

1. factor recovery, measured by cosine matching between fitted rows of F and true rows of F_0 ;
2. motif clustering accuracy, measured by matching $\arg \max_s \omega_{is}$ to true groups;
3. support recovery, measured by converting posterior motif scores to predicted active factors;
4. soft clustering diagnostics, based on group-level averages of ω_{is} rather than only hard labels;
5. posterior uncertainty summaries for $\bar{\gamma}_{sd}$, including entropy, top-two gaps, and targeted pairwise split metrics;
6. selected motif dimensions $\hat{D}_s$ from the ARD prior means;
7. predictive fit, such as held-out Poisson log likelihood or deviance.

5.3 Core simulation settings

The current roadmap suggests the following simulation studies.

Correctly specified block motifs. Use correct K and nominal D , with groups occupying simple single-factor and multi-factor supports. This verifies that the method can recover factors, clusters, and supports in a clean setting. Current evidence shows feasibility, but a larger seed grid is needed before claiming robust superiority over baselines.

Tree/no-anchor motifs. Use motifs such as $(1, 2, 3)$, $(1, 2, 4)$, $(1, 5, 6)$, $(1, 5, 7)$, $(1, 2)$, and $(1, 5)$, so that no group is explained by a single anchor factor. This is the strongest current positive setting: in a small multi-seed pilot, MoS-NMF has substantially better support recovery than the baseline pipelines.

Overspecified K . Generate data with K true factors but fit with extra candidate factor rows, including near-duplicates of true factors. The key diagnostic is whether $\bar{\gamma}_{sd}$ splits posterior mass across duplicate rows when they are statistically interchangeable. Global entropy is not sufficient; the paper should report pair-specific posterior split metrics for the duplicated factors.

Correlated factors. Generate factor rows with controlled correlation strength. This tests whether MoS-NMF can express targeted ambiguity between correlated factors. The current evidence suggests that raw entropy can be misleading because weak dimensions may become globally diffuse. The JMLR version should sweep correlation strength and report targeted pair uncertainty, not only mean entropy.

Overspecified D . Fit each motif with a nominal dimension larger than the true motif dimension. The target behavior is that extra dimensions either have small ARD prior mean $\alpha_{sd}^0/\beta_{sd}^0$ or small posterior loading mass. The current single-seed evidence suggests that diffuse-surplus initialization helps, while repeated-surplus initialization duplicates active dimensions. A full paper needs multi-seed threshold sweeps for $\hat{D}_s$ and comparisons against the true motif dimensions after matching fitted motifs to groups.

5.4 Real data

A JMLR-quality paper should include at least one real-data example where repeated sparse loading motifs have scientific meaning. The analysis should report the fitted factors, the motifs, motif-specific selected dimensions, sample responsibilities, and posterior uncertainty over ambiguous motif dimensions. The real-data example should be chosen so that the learned motifs can be interpreted externally, not only through reconstruction quality.

5.5 Ablations and robustness checks

The paper should include ablations for:

1. fixed versus learned F ;
2. initialization from Poisson NMF versus MF-Poisson-SuSiE;
3. sensitivity to the number of motifs S ;

4. sensitivity to nominal D and the ARD threshold τ ;
5. sensitivity to duplicate-factor and correlated-factor strength;
6. stability across random initializations.

These experiments are needed to distinguish the proposed statistical model from an initialization artifact.

6 Discussion

MoS-NMF reframes sparse NMF as a motif discovery problem. Instead of estimating sparse loadings independently for each observation, it pools repeated sparsity patterns through latent motifs. The posterior motif probabilities $\bar{\gamma}_{sdk}$ provide an interpretable uncertainty summary over the factor identities defining each motif, while the responsibilities ω_{is} cluster observations by motif. The empirical-Bayes prior means $\alpha_{sd}^0/\beta_{sd}^0$ provide a natural ARD-style diagnostic for the effective dimension of each motif.

The current evidence supports the promise of this approach, especially in tree/no-anchor simulations and in overfitted-factor uncertainty diagnostics. The main open issues are stable dimension selection, broader multi-seed benchmarks, model-selection guidance for K , S , and D , and a real-data example with scientifically interpretable motifs.

References

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